

Project design phase Technology Stack (Architecture & Stack)

Date	1 july 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	4 Marks

Technical Architecture:

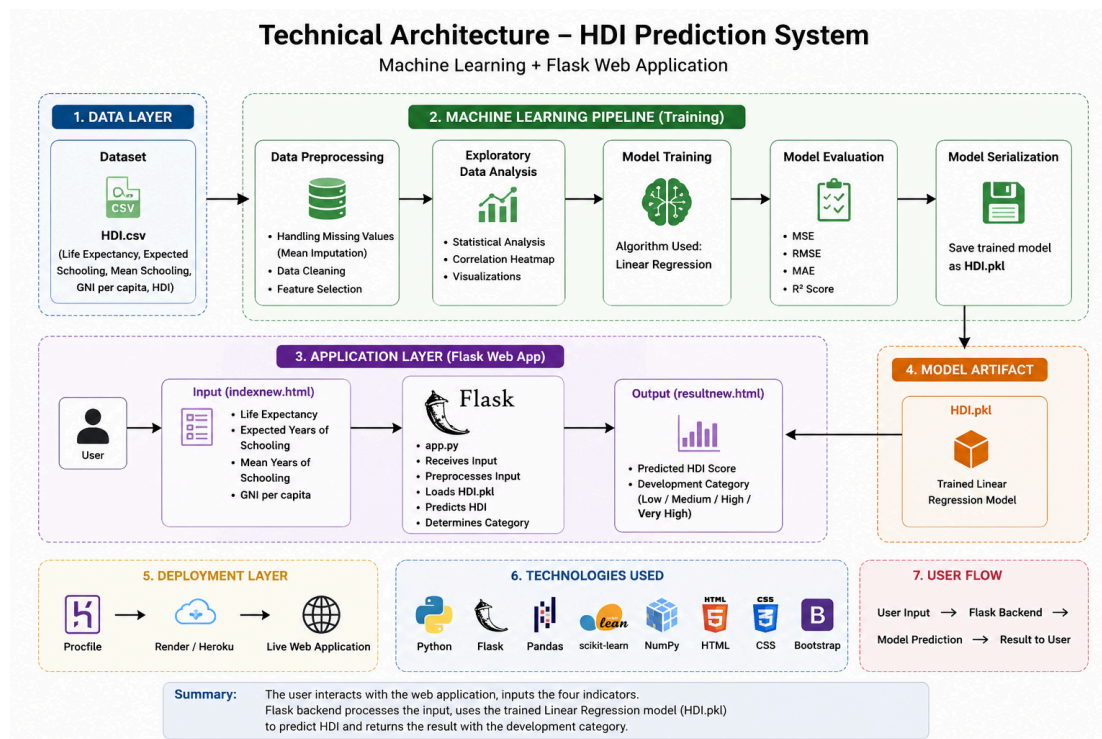


Table-1 : Components & Technologies:

s.no	component	description	Technology used
1	User Interface	Web interface where users enter Life Expectancy, Expected Years of	HTML, CSS, Bootstrap

		Schooling, Mean Years of Schooling, and GNI per capita to predict HDI.	
2	Application Logic-1	Accepts user inputs, validates the values, and sends them to the prediction model.	Python, Flask
3	Application Logic-2	Loads the trained machine learning model (HDI.pkl) and processes the input data for prediction.	Python, Pickle, scikit-learn
4	Application Logic-3	Predicts the HDI score and categorizes it into Low, Medium, High, or Very High Human Development.	Linear Regression (scikit-learn)
5	Database	Dataset containing HDI indicators used for model training. No runtime database is used in the application.	CSV File (HDI.csv), Pandas
6	Cloud Database	Not applicable. The project stores data locally and does not use a cloud database.	N/A
7	File Storage	Stores the trained machine learning model, dataset, HTML templates, and static files.	Local File System (HDI.pkl , CSV, Templates)
8	External API-1	No external APIs are used in the project.	N/A
9	External API-2		N/A

		No external APIs are integrated.	
10	Machine Learning Model	Predicts the Human Development Index (HDI) score based on socioeconomic indicators.	Linear Regression, scikit-learn
11	Infrastructure (Server/Cloud)	Flask application deployed on a local server. Can also be deployed on Render, Railway, or Heroku using the Procfile.	Flask, Python, Local Server / Render

Table-2: Application Characteristics:

s.no	component	description	technology
1	Open source framework	The application is developed using open-source frameworks for machine learning, web development, and data processing.	Flask, scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Bootstrap
2	Security Implementations	Basic input validation is implemented to ensure users enter valid numeric values. The application runs locally and does not store user data or require authentication. HTTPS can be enabled during deployment.	Flask Input Validation, Python Validation, HTTPS (optional for deployment)
3	Scalable Architecture	The application follows a modular architecture consisting of a User Interface, Flask backend, Machine Learning model, and local file storage. It can be deployed on cloud platforms such as Render or Heroku for better scalability.	Flask, Python, Render/Heroku (optional), Pickle (.pkl)
4		The application is available through a Flask web	Flask Development

	Availability	server. It can run continuously on a local machine or be hosted on a cloud platform for 24×7 accessibility.	Server, Render, Heroku, Gunicorn (optional)
5	Performance	The trained Linear Regression model loads quickly and generates HDI predictions within milliseconds. Since the application performs only inference, it requires minimal CPU and memory resources and supports multiple prediction requests efficiently.	Linear Regression (scikit-learn), Flask, Python